CAP6419: 3D Computer Vision
Assignment 03: Panoramic Image Generation
Due Date: 01Nov2025
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Report Panoramic Image Generation

1. Introduction:

In this exercise, a sequence of images was taken and stitched together to create a wide panoramic image by registering, warping and compositing them together. To do this, these general steps were taken:

1. Obtain at least 4 images in a scene where the camera projection center does not change and are overlapping. What this means is the photographer needs to take 4 images by rotating his or her body without moving side to side to take a few pictures with a camera. Each picture has parts of the image that overlap with the next image.
2. Use a feature point correspondence algorithm such as SIFT with RANSAC to automatically track point correspondences. Set the number of features to track (i.e 100). Run the point correspondence algorithm with your image sequence to create a sequence of point correspondences.

3. Compute the infinite homography between each pair of adjacent frames
4. Warp each image through the infinite homography associated with one the middle of the frames using the backward mapping method.
5. Composite all images into a single panoramic image.

2. Description of the methods implemented for accomplishing the panoramic generation:

The panorama generation follows this pipeline:

1. Input Images
2. Cylindrical Warping
3. Feature Matching using SIFT
4. Transformation Estimation using RANSAC
5. Drift Correction
6. Alpha Blending
7. Output Panorama

2a. Cylindrical Warping

The purpose of cylindrical warping is to transform planar images into cylindrical coordinates in order to compensate for the rotation only camera motion. This means that when you take pictures of a scene by rotating around in the same spot you are standing… This section has to geometrically account for that.

2b. Feature Matching using SIFT

The purpose of feature matching is to go into the images and extract distinctive keypoints in the image that are common between the images even if one or both of the images are scaled or rotated differently.

2c. Transformation using RANSAC

The purpose of this is to estimate 2D translation between image pairs while filtering out outlier data. Basically this means when you rotate the camera and take pictures, this algorithm detects and corrects for the movement of your camera and the different part of the image that was captured in the picture.

2d. Drift Correction

This purpose of this section is to then anchor all the image to a common reference frame. This will be the middle image in the series of images that were taken for the panorama.

2e. Alpha Blending and Image Merging

Stitch the images together and smooth the transition between one image and another.

3. Description of the parameter values used and how they were selected:

3a. SIFT Feature Detection Parameters

Parameter	Value	Reason
Edge Threshold	10	Lower threshold detects more features and 10 seemed to be a good balance between not selective enough and too selective

3b. RANSAC Parameters

Parameter	Value	Justification
Confidence	0.99	Probability of finding correct model. High value ensures reliable estimation even with a lot of outliers
Inlier Ratio	0.3	Percentage of Inliers. This value accounts for many incorrect matches from SIFT
N Pairs	1	Translation requires only 1 point pair. Allows efficiency of RANSAC calculations
Epsilon	1.5	Inlier Threshold. Pixels that are similar within 1.5 pixels

3c. Focal Length Parameters

Dataset Type	Focal Length (Pixels)	Reasoning
Phone Cameras	800	Phone Camera FOV (70 degrees)
Wide angle Landscapes	400	Lower Focal Length for wide FOV captures. Reduces warping artifacts
Standard Camera	595	FOV (60-65 degrees)
Zoomed Image	1000-2000	Higher Focal Length for narrow FOV

Results Showing Input Images and the Resultant Panorama:

Figure 2: Input Image 1

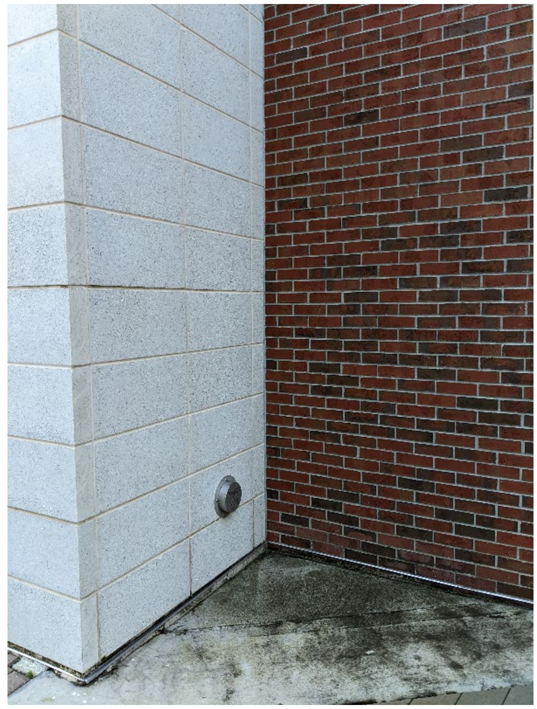

Figure 1: Input Image 2

Figure 3: Output Panorama

Figure 5: Input Image 1

Figure 6: Input Image 2

Figure 4: Output Panorama